

Mortgage Prepayment Risk Modeling

KM & Cox · ML & Deep Learning · Competing Risks · Neural Survival · Scenario Analysis

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May 13, 2026

Goal. Build a survival-analysis pipeline for mortgage prepayment on Freddie Mac single-family data, comparing classical, ML, and deep-learning approaches.

What we implemented:

- **Part A.** Kaplan–Meier survival on the full 31.7M-loan population, stratified by FICO/LTV/vintage/purpose
- **Part B.** Cox proportional hazards (origination + vintage-macro covariates), PH diagnostics
- **Part C.** ML models (LogReg, RF, LightGBM) for horizon-specific prepayment prediction; Cox baseline
- **Part D.** Deep Cox model (DeepSurv) + linear-Cox-via-SGD ablation
- **Part E.** Competing risks (AJ), time-varying Cox (Andersen–Gill), neural survival (LongitudinalDeepHit), scenario analysis

Headline result. Deep Cox: $AUC = 0.729$ at $T=60$ vs 0.685 linear baseline ($\Delta = +0.045$). Gap widens monotonically with horizon.

Data: Freddie Mac Single-Family Loan-Level

Population. 31,688,129 30-year fixed-rate mortgages, originated 1999–2025.

Outcomes (cause-specific framing for prepayment):

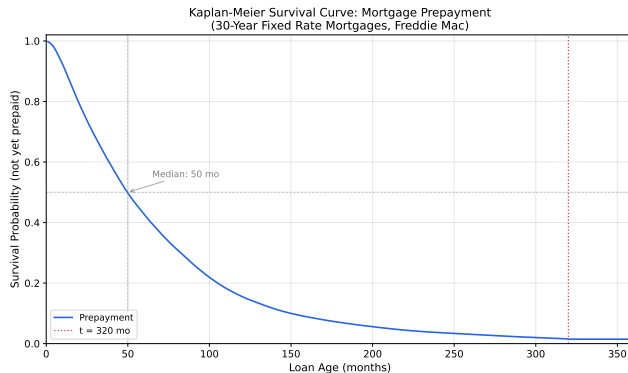
- Prepaid (Zero Balance Code 01): 21,748,774 (68.6%) — the event
- Credit-related termination (ZBC 03, 09): 532,230 (1.7%)
- Other termination (ZBC 02, 15, 16, 96): 265,311 (0.8%)
- Right-censored (still active): 9,141,814 (28.8%)

Survival table. One row per loan with `Duration`, `Event_Prepay`, and 39 origination covariates. Built once via polars, persisted to parquet. Sentinel codes (`DTI=999`, `MI=999`, `Credit=9999`) cleaned to `NaN`.

Train/test split. By vintage — train: ≤ 2014 (full crisis cycle); test: 2015–2019.

Part A — Kaplan–Meier on the Full Population

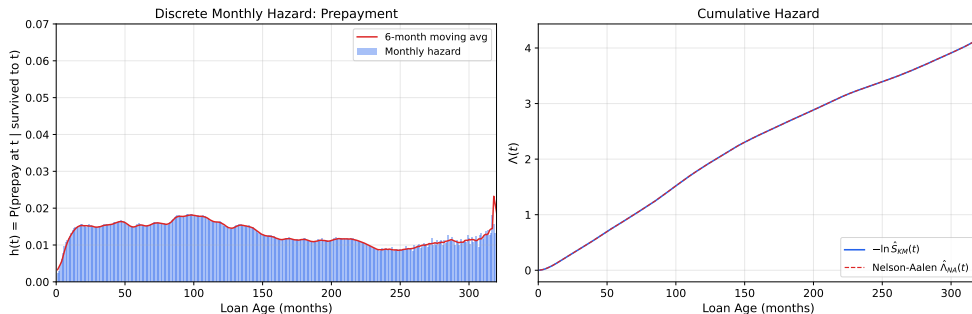
No sampling: KM fit on all 31.7M loans.



All non-prepay outcomes pooled as right-censored (cause-specific framing):
9.94M censored vs 21.75M events.

Part A — Hazards

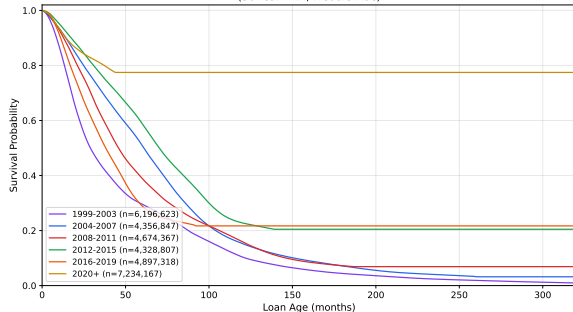
Prepayment Hazard Analysis (30-Year FRM, Freddie Mac)



Discrete monthly hazard: ramps from $\approx 1\%$ at $t=6$ to $\approx 1.4\%$ at $t=12$, then plateaus at 1.5–1.7%.
Sanity check: $-\log S_{KM}$ vs Nelson–Aalen cumulative hazard agree to within numerical precision.

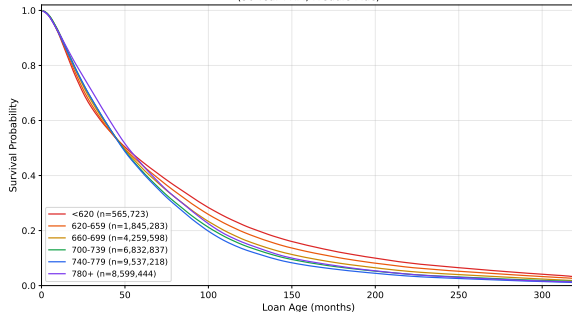
Part A — Stratification by Vintage and FICO

Prepayment Survival by Origination Vintage
(30-Year FRM, Freddie Mac)



Vintage dominates. 1999–2003: 96% prepay, median 29 mo (post-2000 rate cuts). 2020+: only 16% prepay, median undefined (high-rate environment).

Prepayment Survival by FICO Score
(30-Year FRM, Freddie Mac)



FICO is monotone but modest. Prepay rate ranges from 78% (sub-620) to 65% (780+). Higher-FICO borrowers prepay less in proportion.

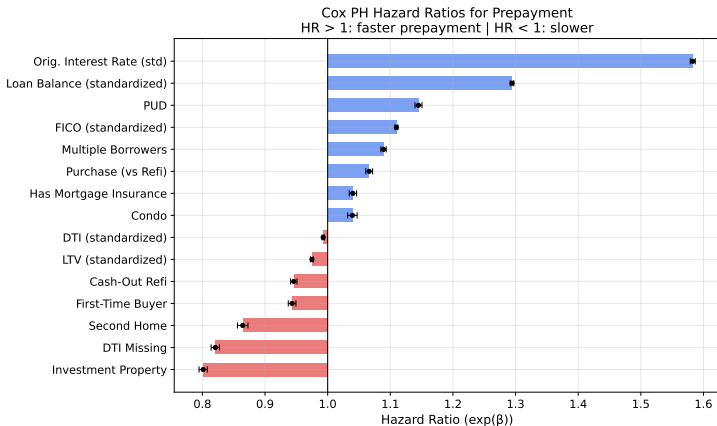
Part B — Cox PH with Origination Covariates

Base model. 5% sample (1.58M loans, 1.08M events), 15 origination covariates, lifelines + ℓ_2 penalizer 0.001.

Concordance index: 0.6257.

Most important hazard ratios:

- Original interest rate: 1.58
- Original UPB: 1.29
- Investment property: 0.80
- DTI missing: 0.82
- FICO: 1.11



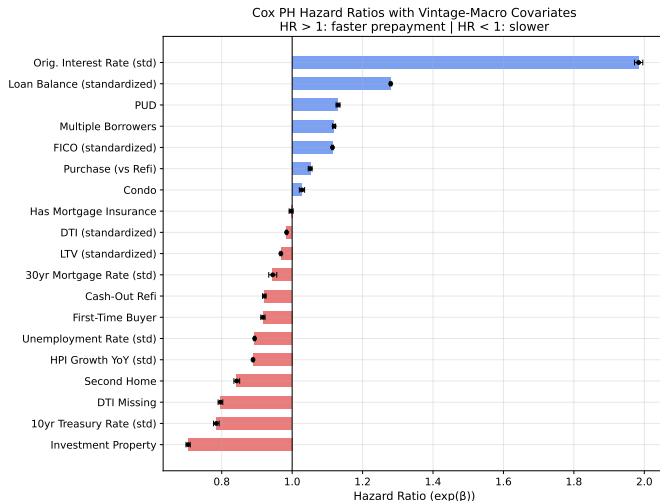
Part B — Vintage-Macro Extension

Macro covariates joined at **origination**. Concordance: 0.6428 ($\Delta C = +0.017$).

Macro hazard ratios:

- 10y Treasury: 0.78
- HPI (seasonally adj.): 0.89
- Unemployment: 0.89
- 30y mortgage rate: 0.94

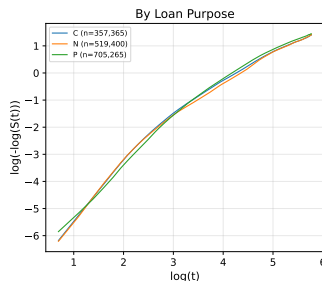
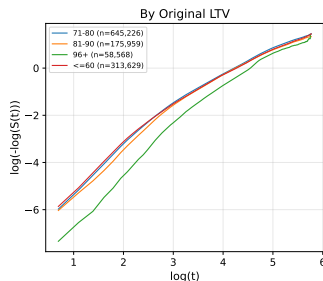
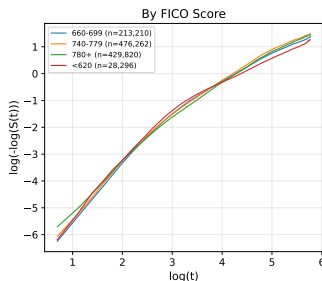
Caveat: vintage-macro covariates capture the macro state *when originated*, not time-varying through the loan's life.



Part B — Proportional Hazards Diagnostic

Visual PH check: log-log Kaplan–Meier curves by FICO, LTV, and loan purpose.

Log-Log Survival Plots
(Parallel curves $\Rightarrow$ PH holds)



- $\log[-\log S(t)]$ vs $\log(t + 1)$; under PH, curves should be parallel
- Strata bend and converge/diverge over time — covariate effects are time-varying, motivating ML/deep models

Part C — ML Models for Horizon-Specific Prediction

Reframing. For each horizon $T \in \{12, 24, 36, 60\}$:

$P(\text{prepaid by } T \mid X)$ as a binary classification.

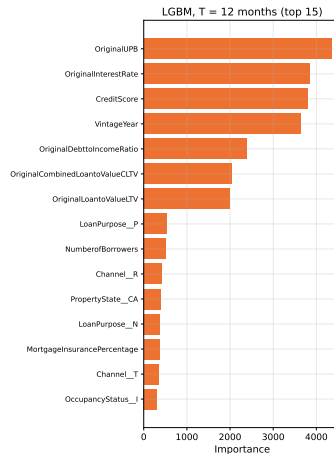
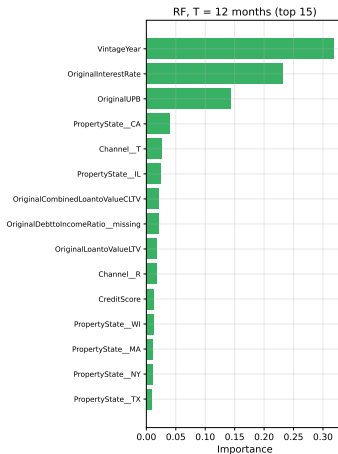
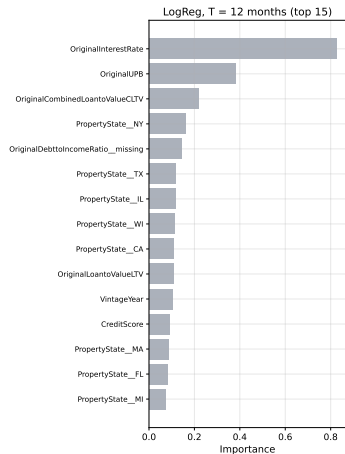
Setup: 10% sample, 1.85M train / 597k test, 68 features after one-hot. Same canonical split as Part D.

Four models:

- Logistic Regression (ℓ_2 , internal scaler)
- Random Forest (200 trees)
- LightGBM (with held-out-validation early stopping)
- Cox baseline (lifelines, predict $1 - S(T \mid X)$)

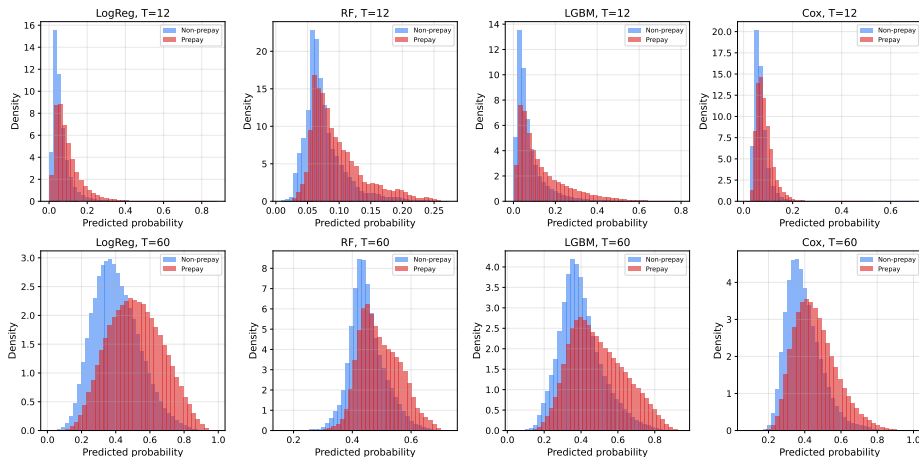
Evaluated: AUC, Brier, log-loss, accuracy — per model, per horizon, stratified by vintage and FICO.

Part C — Feature Importance (T=12)



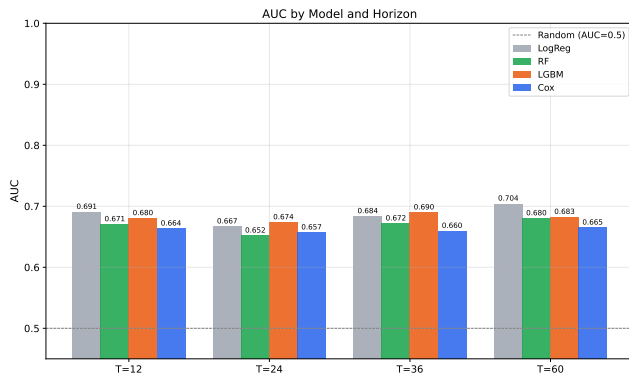
Top 15 predictors for each ML model at the T=12 horizon.

Part C — Score Distributions



Predicted prepayment probabilities at T=12 and T=60, by model and realized prepayment status.

Part C — Out-of-Sample AUC

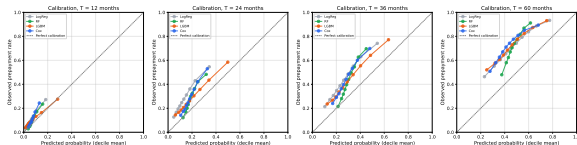


Model	T=12	T=24	T=36	T=60
Cox	0.66	0.66	0.66	0.67
RF	0.67	0.65	0.67	0.68
LGBM	0.68	0.67	0.69	0.68
LogReg	0.69	0.67	0.68	0.70

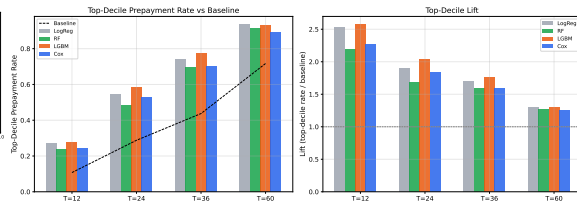
Observations:

- All four models in the 0.65–0.70 band
- LogReg competitive, even best at T=12 and T=60
- Marginal effects largely linear-additive

Part C — Calibration and Top-Decile Lift



Calibration is reasonable for all models.



Top-decile lift. At T=12 the top decile catches 27.6% prepayments (LightGBM) vs 10.75% baseline → **lift** = $2.57\times$.

Model: $\lambda(t | X) = \lambda_0(t) \cdot \exp(f_\theta(X))$ where f_θ is a feed-forward network.

Architecture (DeepSurv-standard):

- 3 hidden layers $\times$ 64 ReLU units
- Batch normalization, dropout 0.1
- 13,184 parameters

Training:

- Adam, $\eta = 10^{-3}$, batch 512
- Up to 50 epochs, ES patience 8
- Breslow baseline post-fit on full train

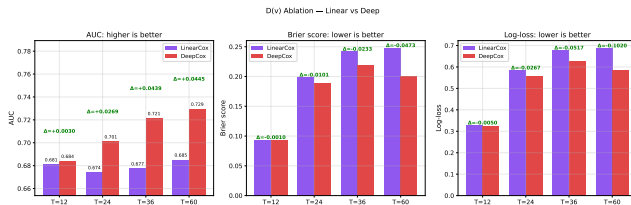
Linear baseline (D(v) ablation):

- Same pycox infrastructure
- `num_nodes=[]` $\rightarrow$ no hidden layers
- 68 parameters; functionally a SGD-fit classical Cox

Why cleanest ablation: same data, optimizer, baseline-hazard estimator — only difference is depth.

Runtime: 17 min Deep + 1 min Linear on CPU.

Part D — Linear vs Deep Ablation ($D(v)$)

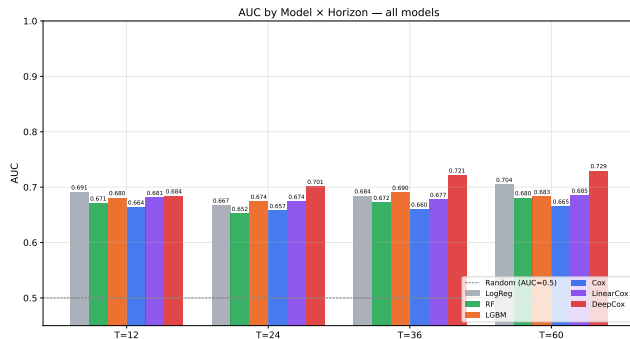


T	Linear	Deep	Δ AUC
12	0.6807	0.6837	+0.003
24	0.6738	0.7007	+0.027
36	0.6772	0.7211	+0.044
60	0.6848	0.7292	+0.045

The gap widens monotonically with horizon.

Sadhwani et al. (2021): nonlinear interactions accumulate over the loan's life.

Part D — Head-to-Head with All Models

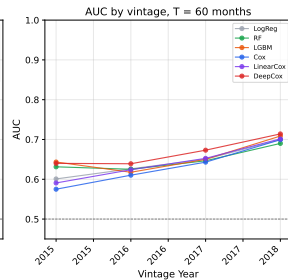
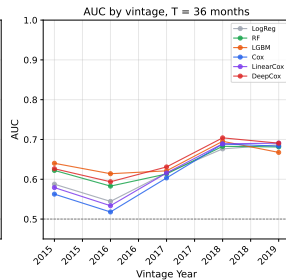
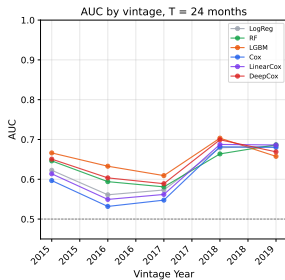
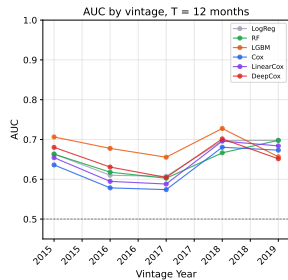


Model	T=12	T=24	T=36	T=60
Cox	.66	.66	.66	.67
RF	.67	.65	.67	.68
LGBM	.68	.67	.69	.68
LinearCox	.68	.67	.68	.68
LogReg	.69	.67	.68	.70
DeepCox	.68	.70	.72	.73

DeepCox is best at every horizon ≥ 24 months.

At T=12 LogReg edges it — short-horizon prepay dominated by easy cases.

Part D — AUC by Vintage



Why Deep Wins at Long Horizons

Linear assumption: log-hazard is additive in covariate effects.

$$\log \lambda(t \mid X) = \log \lambda_0(t) + \beta_1 x_1 + \beta_2 x_2 + \dots$$

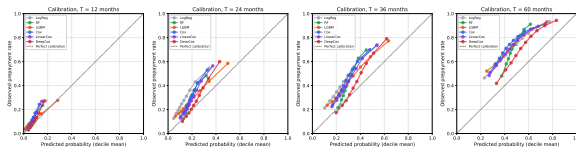
Reality: the refinancing decision at $T=36$ depends on FICO, balance, and rate environment *jointly* — non-trivial interactions.

At long horizons, more interaction paths matter. The deep network captures them; the linear model collapses them. Hence the widening gap.

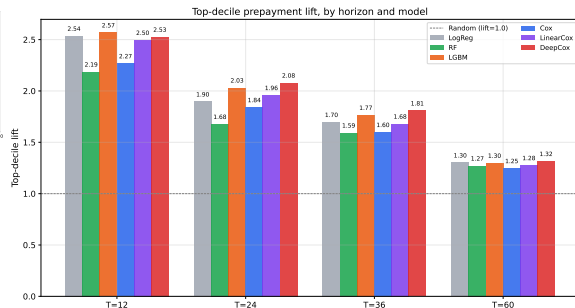
At $T=12$ only borrowers with very strong, singular incentive have prepaid — easy cases for any model.

Same conclusion as Sadhwani–Giesecke–Sirignano (2021) on the much larger CoreLogic dataset.

Part D — Calibration and Top-Decile Lift



Calibration. DeepCox close to diagonal at all horizons. LinearCox slightly under-confident.



Top-decile lift.

- T=24: DeepCox 2.08 $\times$ vs LGBM 2.03 $\times$
- T=36: DeepCox 1.81 $\times$ vs LGBM 1.77 $\times$

Key Takeaways: Parts A–D

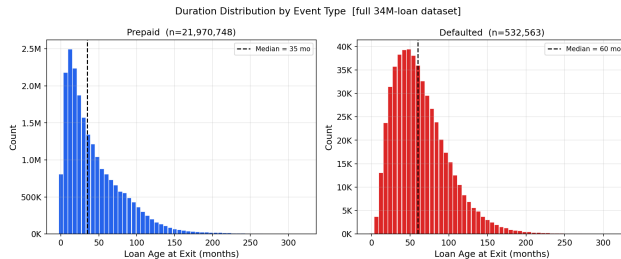
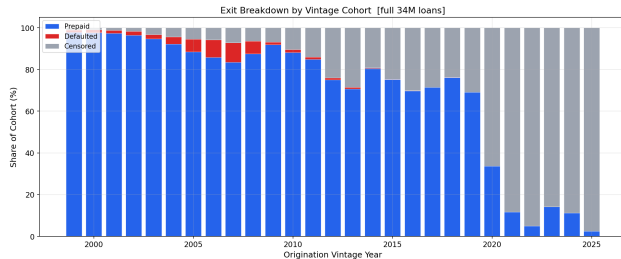
- ➊ **Mortgage prepayment is not a linear-in-covariates process.** Schoenfeld test rejects PH for 12/15 covariates.
- ➋ **Deep Cox AUC = 0.729 at T=60** vs 0.685 linear baseline ($\Delta = +0.045$). Gap grows with horizon.
- ➌ **Tree models don't close the gap.** LightGBM tops out at 0.68–0.69. Interactions are smooth-and-multivariate (good for MLPs) rather than coarse-and-axis-aligned.
- ➍ **Logistic regression is competitive at T=12.** When prepayment concentrates in an obvious refi tail, marginal effects suffice.
- ➎ **Vintage-macro covariates earn their place.** Cox concordance jumps from 0.626 to 0.643 from FRED data alone.

Part E(i)

Competing Risks: Prepayment vs. Default

Kaplan–Meier vs. Aalen–Johansen · Cause-Specific Cox · Fine–Gray

Prepaid and default are competing risks.



Kaplan–Meier vs. Aalen–Johansen.

Kaplan–Meier

$$\hat{F}_{\text{KM}}^{(k)}(t) = 1 - \prod_{t_j \leq t} \left(1 - \frac{d_j^{(k)}}{n_j^{\text{KM}}} \right)$$

$d_j^{(k)}$ cause- k exits at t_j
 n_j^{KM} risk set excluding competing-event loans

Aalen–Johansen

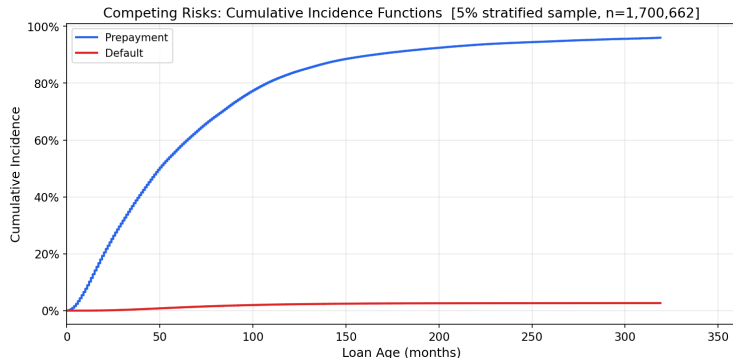
$$\hat{F}_k(t) = \sum_{t_j \leq t} \hat{S}(t_j^-) \cdot \frac{d_{kj}}{n_j}$$

d_{kj} cause- k exits at t_j
 n_j risk set including all exits
 $\hat{S}(t_j^-)$ overall survival just before t_j

KM: competing exits censored $\Rightarrow$ denominator too small $\Rightarrow$ each increment inflated.

AJ: $\hat{S}(t_j^-)$ shrinks with every exit $\Rightarrow \hat{F}^{(\text{pre})} + \hat{F}^{(\text{def})} + \hat{S} = 1$ always.

Aalen–Johansen CIF: coherent probability partition.



Cumulative Incidence Function

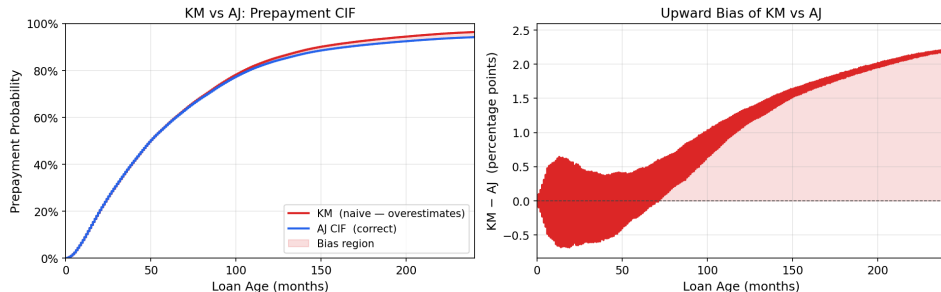
$$F_k(t) = P(\text{exit by } t, \text{ cause } k)$$

Aalen–Johansen CIF partition

	Prepaid	Default	Active
5 yr	61.4%	1.1%	37.6%
10 yr	83.3%	2.2%	14.5%
20 yr	91.8%	3.1%	5.1%

KM overstates prepayment by up to 2.2 pp. over 20 years.

KM over-estimates prepayment CIF by treating defaults as random censoring

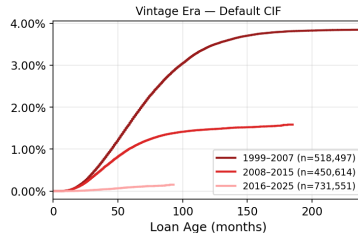
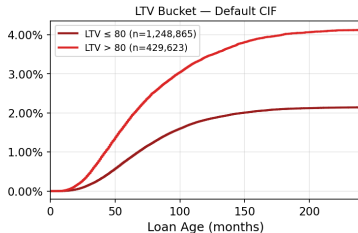
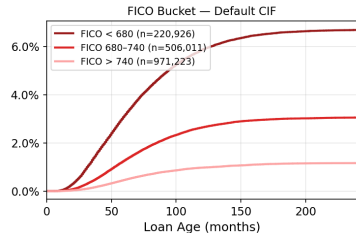
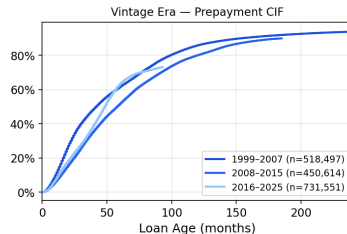
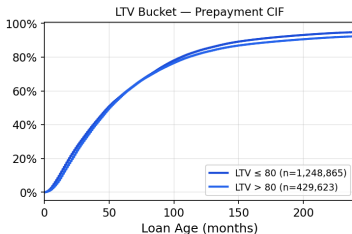
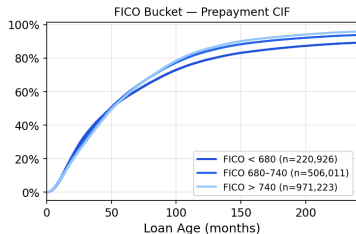


KM overstatement of prepayment probability

Horizon	$\hat{F}_{KM}$	$\hat{F}_{AJ}$	Bias
5 yr	61.9%	61.4%	+0.45 pp
10 yr	84.6%	83.3%	+1.31 pp
20 yr	93.4%	91.8%	+2.23 pp

FICO and LTV separate default, not prepayment.

Stratified Competing-Risk CIFs [5% sample]



Cause-specific Cox model.

Cause-Specific Cox

$$h_k(t | \mathbf{x}) = h_{0k}(t) \exp(\boldsymbol{\beta}_k^\top \mathbf{x}), \quad k \in \{\text{pre}, \text{def}\}$$

$$\ell(\boldsymbol{\beta}_k) = \sum_{i: E_i=k} \left[\boldsymbol{\beta}_k^\top \mathbf{x}_i - \log \sum_{j \in \mathcal{R}(t_i)} e^{\boldsymbol{\beta}_k^\top \mathbf{x}_j} \right]$$

$\mathcal{R}(t_i)$: all loans still active just before t_i .

Competing-cause exits are **censored** — they leave $\mathcal{R}(t)$ without contributing to ℓ .

Additional features

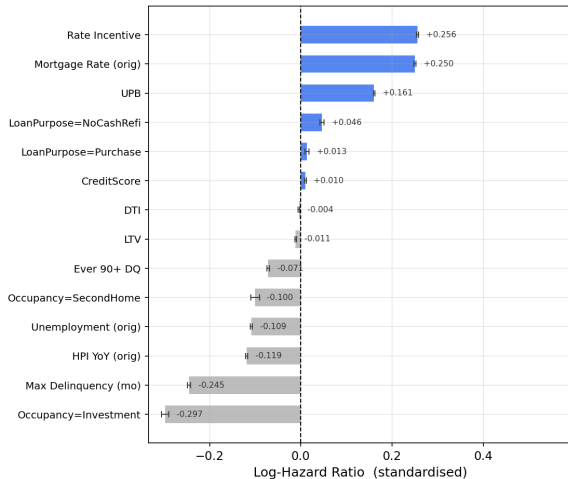
Max Delinquency — months past due over loan life

Ever 90+ DQ — binary flag for severe delinquency

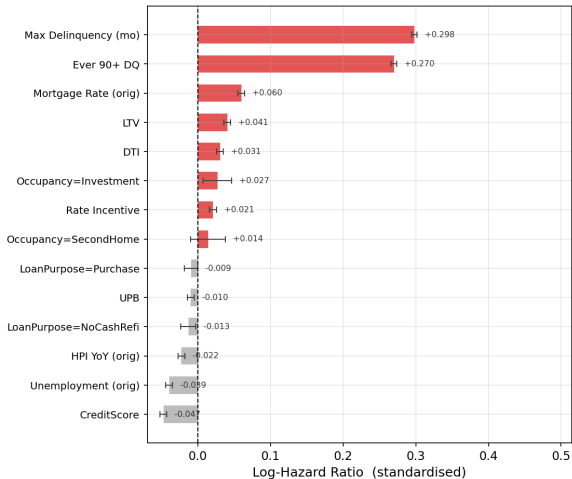
Opposite covariate effects by cause.

Prepayment and default respond to opposite covariate effects

Cause-Specific Cox — Prepayment



Cause-Specific Cox — Default



Four covariates flip sign across causes.

Naive Cox vs. cause-specific coefficients

Covariate	Naive $\hat{\beta}$	Prepay $\hat{\beta}$	Default $\hat{\beta}$	
Rate incentive	+0.253	+0.256	+0.021	(pure prepay)
UPB	+0.161	+0.161	-0.010	(pure prepay)
FICO	+0.016	+0.010	-0.047	opposite
LTV	-0.007	-0.011	+0.041	opposite
Max Delinquency	-0.123	-0.245	+0.298	opposite
Ever 90+ DQ	-0.035	-0.071	+0.270	opposite

Each extra month delinquent raises default hazard by 35% and cuts prepayment hazard by 22% — yet the naive coefficient (-0.123) conceals both, collapsing a $+0.543$ spread to near zero.

$$e^{+0.298} - 1 = +35\% \text{ (default); } 1 - e^{-0.245} = +22\% \text{ (prepay reduction); } \text{spread} = 0.298 - (-0.245) = 0.543.$$

Key Takeaways: E(i)

- ① Default and prepayment are competing risks — a naive Cox model conflates both.
- ② **Kaplan–Meier overstates prepayment by up to +2.2 pp over 20 years; use Aalen–Johansen.**
- ③ Vintage era dominates: 2007 cohort defaults at $6\times$ the long-run average.
- ④ FICO drives default; rate incentive drives prepayment — separate channels.

Part E(ii)

Time-Varying Covariates in Mortgage Survival

Andersen–Gill Cox Extension · Live Rate Incentive · Landmark C-index

Static Cox vs. Andersen–Gill.

Standard Cox (11 features)

One row per loan; all covariates fixed at origination.
Cannot track rate cycles, burnout, or equity build after origination.

Andersen–Gill (13 features)

One row per loan-month; $x_i(t)$ updated each period.
Extends the same static base with four time-varying features. 100K loans $\times$ ~ 47 months = 4.68M intervals.

Time-varying features — updated every month

Feature	What it captures
rate_incentive	$r_{\text{orig}} - r_{\text{mkt}}(t)$: live option value to refinance
ELTV	equity from amortisation and HPI gains
unemployment	macro labour cycle
hpi_yoy	home-price appreciation unlocking cash-out refi

*unemployment and hpi_yoy also in Standard Cox as origination-month snapshots.

The Andersen–Gill model.

Hazard function (identical in form to standard Cox)

$$h_i(t) = h_0(t) \exp(\beta^\top x_i(t))$$

$x_i(t)$ is the **current** covariate vector, updated each month.

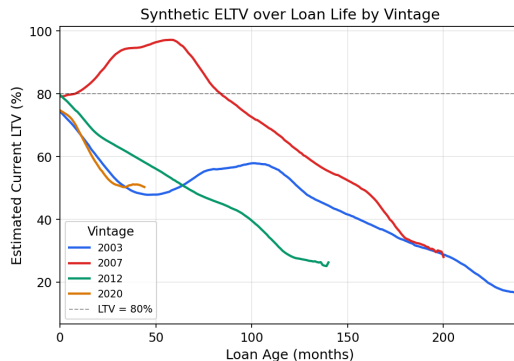
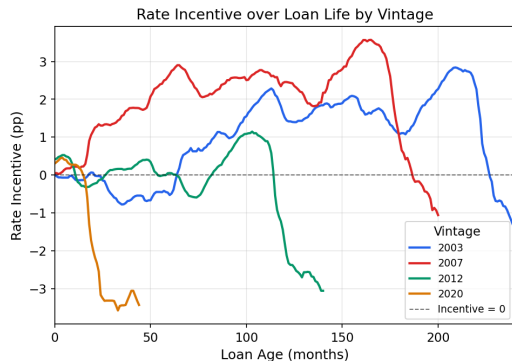
Log partial likelihood

$$\ell(\beta) = \sum_{i: \delta_i=1} \left[\beta^\top x_i(t_i) - \log \sum_{j \in \mathcal{R}(t_i)} \exp(\beta^\top x_j(t_i)) \right]$$

At each event: *given a prepayment at t_i , what is the probability it was loan i rather than any other active loan?*

Time-varying feature trajectories by vintage.

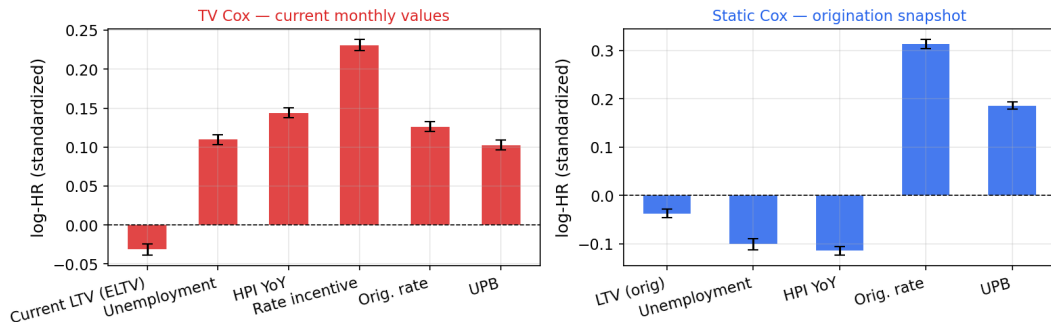
Static Cox uses only month-0 values — Andersen-Gill tracks these every month



Static Cox uses only month-0 values — Andersen-Gill tracks these every month.

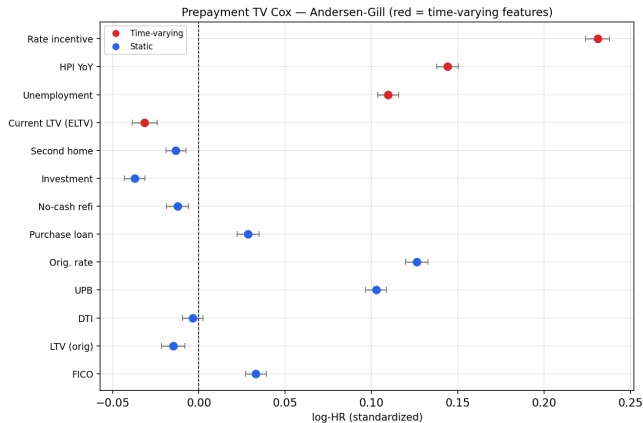
Static vs. TV Cox coefficient comparison.

Feature coefficients: TV Cox (current monthly) vs Static Cox (origination snapshot)
Shared static features (Orig. rate, UPB) shown in both panels



- `rate_incentive`: dominant positive predictor in TV Cox; near zero at origination, absent from Static Cox
- Macro signals (unemployment, HPI) flip sign — vintage-era snapshots capture a different regime than monthly values

Landmark C-index: TV Cox leads to $T=60$, then reverses.



Landmark C-index

$$C(t^*) = P(\hat{h}_i(t^*) > \hat{h}_j(t^*) \mid T_i \leq t^* < T_j)$$

Horizon	Static	TV	Δ
$T = 12$	0.609	0.681	+0.072
$T = 24$	0.627	0.687	+0.060
$T = 36$	0.630	0.675	+0.045
$T = 60$	0.625	0.646	+0.021
$T = 84$	0.623	0.605	-0.018
$T = 120$	0.620	0.583	-0.037

Key Takeaways: E(ii)

- ① Monthly-updated covariates add real information (+1,996 log-likelihood gain).
- ② **TV Cox leads at short/medium horizons: +0.072 at $T=12$, +0.021 at $T=60$.**
- ③ **TV Cox degrades beyond $T \approx 70$ months — use a landmark super-model.**
- ④ Rate incentive is the key driver; macro coefficients require careful inspection.

Part E(iii)

Neural Survival Models: LongitudinalDeepHit

Transformer Encoder · Competing-Risk CIFs · Gradient Sensitivity

Why go beyond Cox?

Cox / Andersen-Gill limitations

- Proportional hazards: covariate effect is **constant over time**
- Single scalar score — cannot model the **S-curve shape** of prepayment
- Competing risks require separate model fits; no shared representation
- Cannot exploit the **sequence structure** of monthly covariate history

What a sequence model can do

- Read the full monthly path $[x_1, \dots, x_T]$ — **burnout is encoded**
- Produce nonlinear, time-varying hazard rates $\lambda_k(t \mid x)$
- Model prepayment and default **jointly** via shared encoder
- Attention weights identify **which months** drive the prediction

Alternatives rejected

- **LSTM** — vanishing gradients at $T=120$; attends only to adjacent months; hard to interpret
- **Single-context head** — near-constant hazard curve; gradient flows only through last hidden state
- **Two separate models** — violates independent censoring; loses shared macro signal

Why this combination

- **Causal Transformer** — reaches any past month in one attention step; weights are interpretable
- **Per-timestep head** — $\lambda(t)=\text{head}(h_t)$: hazard conditioned on full causal history $x_1 \dots x_t$
- **Joint encoder** — chain product enforces $\text{CIF}_1 + \text{CIF}_2 + S = 1$

LongitudinalDeepHit architecture.

Architecture

Input	$(B, T_{\max}=120, D=13)$ padded sequences
Proj.	Linear $D \rightarrow d_{\text{model}}=64$
PE	Sinusoidal positional encoding on loan age
Enc.	$\times 2$ pre-norm layers, 4 heads, causal mask
Head 1	FC $64 \rightarrow 64 \rightarrow 1$ per h_t , Sigmoid $\Rightarrow \lambda_1(t)$
Head 2	FC $64 \rightarrow 64 \rightarrow 1$ per h_t , Sigmoid $\Rightarrow \lambda_2(t)$
CIFs	Competing-risk chain product

Competing-risk CIFs

$$\begin{aligned}\text{CIF}_k(t) &= \sum_{s=1}^t \lambda_k(s) \cdot S(s-1) \\ S(t) &= \prod_{s=1}^t (1 - \lambda_1(s))(1 - \lambda_2(s))\end{aligned}$$

Guarantees $\text{CIF}_1 + \text{CIF}_2 + S = 1$ at every t .

Training objective

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \alpha \mathcal{L}_{\text{rank}}, \quad \alpha = 0.2$$

NLL: discrete-time cause-specific likelihood

Ranking: soft pairwise concordance (DeepHit-style)

Params	76,290
Data	101K loans, 4.5M rows
Epochs	41 (best ep. 29)
LR	3×10^{-4} , cosine

Data treatment

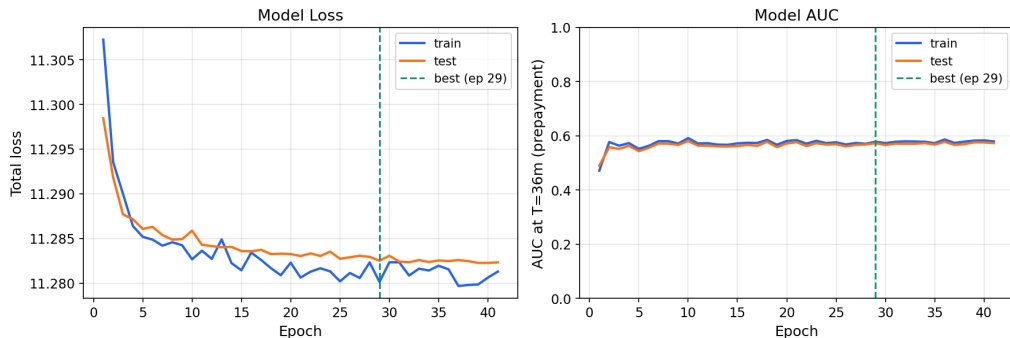
Each loan $\rightarrow$ 13-feature monthly sequence, zero-padded to $T_{\max}=120$ after exit.

Labels: prepayment = 1, default = 2, censored = 0.

Causal mask; 80/20 train-test split.

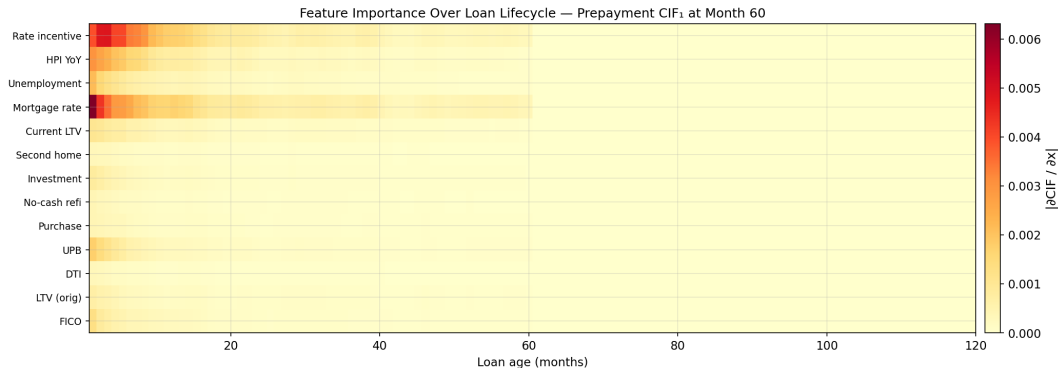
Training dynamics: NLL collapse and ranking loss.

LongitudinalDeepHit — Training Curves



- **NLL freezes:** 1.6% event rate means predicting near-zero hazard for everyone already minimises the loss — rare-event gradients too weak to improve further.
- **Cox immune:** partial likelihood only ranks loans at event times, never predicts absolute hazard — no trivial solution under class imbalance.
- **AUC preserved:** ranking loss descends steadily, ordering loans by relative risk despite collapsed hazard values.

Feature importance over the loan lifecycle.



`rate_incentive` dominates, especially in early months; current LTV rises later as equity accumulates.

Model performance vs. DeepCox.

Prepayment CIF₁ — AUC at fixed horizons

Model	T=12	T=24	T=36	T=60
Part D DeepCox [†]	0.684	0.701	0.721	0.729
DeepHit (20K) [‡]	0.641	0.622	0.572	0.515

Default CIF₂ — AUC at fixed horizons

Model	T=12	T=24	T=36	T=60
Part D DeepCox [†]	0.684	0.701	0.721	0.729
DeepHit (20K) [‡]	0.687	0.708	0.707	0.726

[†] 597K canonical [‡] 20K hold-out **green = better**

Prepayment: DeepCox leads at all horizons (NLL collapse hurts long-horizon calibration).

Default: DeepHit leads at T=12 and T=24 with **6×** **fewer** training loans.

Key Takeaways: E(iii)

- ① Default AUC matches DeepCox with $6\times$ fewer training loans.
- ② Feature importance shifts over the loan lifecycle — unique to sequence models.
- ③ **Limitation: 1.6% event rate causes NLL collapse — AUC is preserved but absolute probabilities are unreliable; fix via focal-loss upweighting.**

Part E(iv)

Scenario Analysis: MBS Prepayment under Rate Shocks

TV Cox Rate Channel · Survival Curves · Monte Carlo MBS Pricing

TV Cox: direct rate channel for scenario analysis.

Why TV Cox (E(ii)) is the primary model

$\text{rate_incentive} = r_{\text{orig}} - r_{\text{market}}(t)$ is a **direct time-varying covariate**.

A Δr bp market rate shock immediately shifts the incentive by $-\Delta r/100$ pp, giving an analytic hazard response:

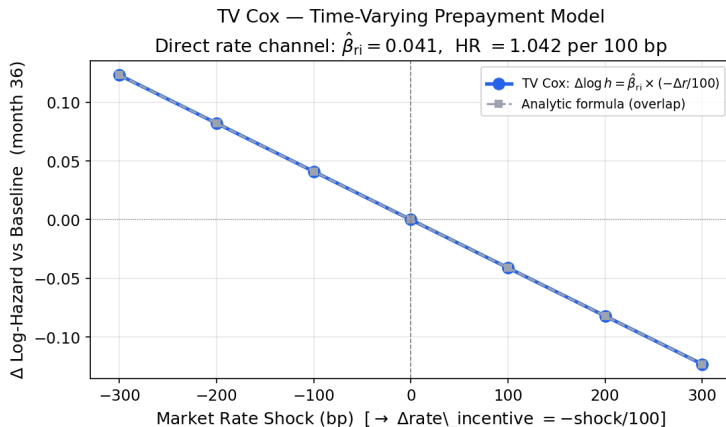
$$\Delta \log h(t) = \hat{\beta}_{\text{ri}} \times \frac{-\Delta r}{100}, \quad \hat{\beta}_{\text{ri}} = 0.041 \text{ (HR} = 1.042 \text{ per 100 bp)}$$

The calibrated baseline hazard $\hat{H}_0(t)$ produces realistic $S(t)$, CPR, WAL, and price.

Representative loan: \$300K, 5.0% coupon, baseline market rate 5.3% (median, 1999–2025).

Scenarios: ± 100 , ± 200 , ± 300 bp — covering the BCBS standard stress range.

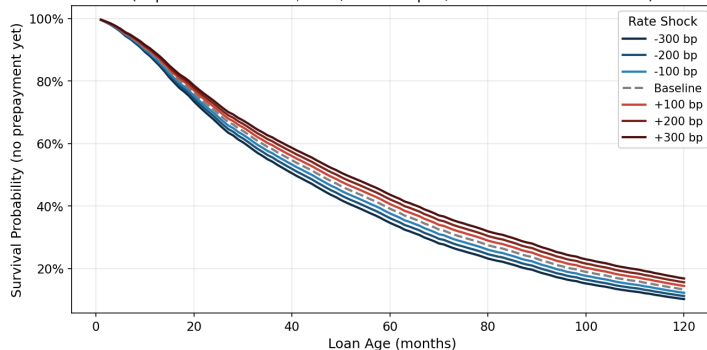
Shock sensitivity: TV Cox matches the analytic formula.



Each 100 bp rate cut increases prepayment hazard by 4.2%; a 300 bp cut raises it by +0.123 in log terms.
Model response matches the closed-form prediction exactly — no simulation needed, fully auditable.

Survival curves under seven rate scenarios.

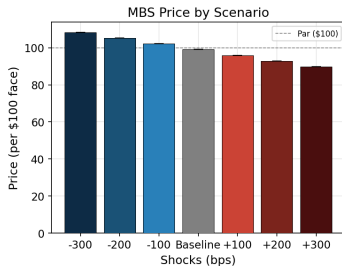
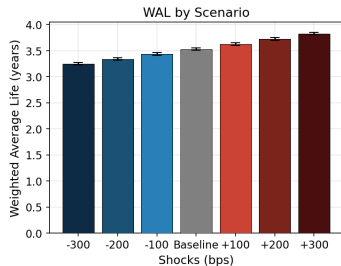
Prepayment Survival Curves Under Interest Rate Scenarios (TV Cox)
(Representative loan: \$300K, 6.5% coupon, baseline market rate 7.0%)



Scenario	10-yr Survival	Median Prepay (mo.)
–300 bp	10.2%	41
–200 bp	11.2%	43
–100 bp	12.2%	45
Baseline	13.3%	46
+100 bp	14.4%	48
+200 bp	15.6%	50
+300 bp	16.8%	51

MBS: Weighted Average Life, price, and negative convexity.

MBS Cash Flow Projection — TV Cox Monte Carlo ($\sigma=25$ bp, 200 paths, 10-yr horizon)
5.0% coupon, \$300K loan, baseline market rate 5.3%



Scenario	WAL (yr)	Price (\$)
−300 bp	3.25	108.32
−200 bp	3.34	105.26
−100 bp	3.43	102.17
Baseline	3.53	99.07
+100 bp	3.63	95.96
+200 bp	3.73	92.86
+300 bp	3.83	89.77

Negative convexity: when rates fall, borrowers prepay at par, capping price upside. Gain (−300 bp: \$9.25) < loss (+300 bp: \$9.30).

MC: 200 paths, $\sigma=25$ bp.

Key Takeaways: E(iv)

- ❶ **Static models give the wrong direction** — time-varying covariates are required for rate scenario analysis.
- ❷ **TV Cox has a linear rate channel**: every 100 bp rate cut adds the same +4.2% to prepayment hazard — equal steps in rate produce equal steps in response, making results predictable and fully explainable.
- ❸ **Negative convexity confirmed**: when rates fall, borrowers prepay at par, capping price upside (\$9.25 gain < \$9.30 loss at ± 300 bp).
- ❹ **FICO and LTV effects are real but require more data to surface** — rate incentive dominates in this specification.

Thank you

Questions?

References

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Katzman et al. (2018), DeepSurv, *BMC Medical Research Methodology*
Kvamme et al. (2019), pycox, *JMLR*
Andersen & Gill (1982), *Ann. Stat.* · Fine & Gray (1999), *JASA*
Lee et al. (2018), DeepHit · Davidson-Pilon (2019), lifelines, *JOSS*
BCBS (2019)

Appendix: Sampling Summary by Part

Part	Sample	Train / Test	Notes
A	Full 31.7M loans	N/A	No sampling; KM fit on entire population
B	5% $\approx$ 1.58M loans	Vintage split (≤ 2014 / 2015–2019)	1.08M events; PH diagnostic on 158K subsample
C	10% $\approx$ 2.45M loans	1.85M train / 597K test	68 one-hot features; same canonical split as D
D	Same 10% as C	1.66M inner train / 185K val / 597K test	Extra 10% val carve for early stopping
E(i)	Full survival table	Same vintage split	Competing-risk labels: prepay=1, default=2, censored=0
E(ii)	100K panel loans	Same vintage split	~ 47 mo avg = 4.68M loan-month intervals
E(iii)	101K panel loans	80K train / 21K test (stratified)	60K prepay + 31K default + 35K censored; drawn from 2M panel loans with monthly history; 4.5M rows
E(iv)	Same as E(ii)	Same as E(ii)	TV Cox model re-used; 200-path Monte Carlo for MBS pricing